ALARGAM MOHAMED YAGOUB OSMAN

AI Engineer | Computer Vision · Edge AI · Intelligent Robotics

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PROFESSIONAL SUMMARY

AI Engineer specializing in Computer Vision, Edge AI, and Intelligent Robotics, with hands-on experience designing, developing, and deploying end-to-end AI systems for real-world applications. Proven ability to build real-time computer vision pipelines, autonomous robotic systems, and edge AI solutions using modern deep learning frameworks. Skilled in optimizing AI models for embedded platforms such as Raspberry Pi and ESP32, and integrating AI with robotics and embedded systems to deliver intelligent, low-latency solutions.

WORK EXPERIENCE

Lead Robotics Engineer (Graduation Project) | Falcons Team Oct 2025 – Jul 2026

- Led the design and development of an autonomous mobile robot (AMR) on ROS 2, achieving real-time mapping and localization in dynamic environments
- Engineered path planning and obstacle avoidance algorithms, reducing navigation errors and improving system response time
- Built a multi-sensor fusion pipeline integrating LiDAR and camera data for enhanced perception
- Deployed PyTorch deep learning models on Raspberry Pi with optimized real-time edge inference

AI Training Specialist (Freelance) | *Outlier* Sep 2024 – Mar 2025

- Validated technical accuracy of AI-generated Python and C++ code, strengthening training data quality for large language models (LLMs)
- Improved model performance through RLHF (Reinforcement Learning from Human Feedback) and high-quality data labeling
- Optimized engineering-focused prompts to improve LLM reasoning and response quality.

Machine Learning Intern | *Amit Learning* Feb 2024 – Apr 2024

- Built machine learning models in Python and TensorFlow, applying advanced data preprocessing techniques
- Deployed predictive AI models to cloud environments, ensuring scalability
- Partnered with cross-functional teams to deliver real-world AI solutions

PROJECTS

ANIMO – AI-Powered Prosthetic Hand | *View Project*

- Developed an AI-powered prosthetic hand combining Computer Vision, Embedded AI, and EMG signal processing for intelligent real-time assistance.
- Implemented an adaptive gesture recognition pipeline to translate muscle activity into natural prosthetic hand movements.
- Integrated embedded hardware, AI inference, and real-time control into a scalable assistive robotics platform.
- Engineered an end-to-end intelligent robotic system optimized for low-latency human-machine interaction.

Delivero – Autonomous Mobile Robot | *View Project*

- Built a full autonomous mobile robot using ROS 2 with SLAM, navigation, and sensor fusion (LiDAR + IMU + encoders)
- Developed a real-time object detection pipeline using YOLO, optimized with TensorRT
- Achieved 12–18 FPS on Raspberry Pi 5 (edge deployment)
- Integrated embedded control using ESP32 and micro-ROS for real-time actuation
- Designed a scalable, modular architecture separating perception, control, and hardware layers

Medical Image Super-Resolution using GAN | *View Project*

- Implemented and trained a GAN-based model for image super-resolution
- Achieved PSNR = 36.1 dB and SSIM = 0.81
- Designed a custom dataset pipeline and preprocessing workflow
- Built generator and discriminator architectures using PyTorch

Real-Time Face Recognition Attendance System | *View Project*

- Developed a real-time face recognition system using OpenCV and the LBPH algorithm
- Built an automated attendance logging system with real-time CSV/Excel reporting
- Designed a secure, admin-controlled model training system
- Implemented a webcam-based real-time detection and recognition pipeline with a Tkinter GUI

Face Mask Detection System | *View Project*

- Developed a CNN-based model for face mask detection, achieving ~92% accuracy on a real-world dataset
- Implemented a full training and inference pipeline using TensorFlow, structured for easy retraining and deployment

PUBLICATIONS / RESEARCH

Autonomous Delivery Robot — AI, Embedded Systems & Robotics | *Show publication*

- Co-authored a research paper on a fully autonomous delivery robot integrating AI, embedded systems, and robotics
- Developed ROS 2 system integration and a micro-ROS bridge between Raspberry Pi 5 and ESP32
- Implemented real-time sensor aggregation and multi-modal sensor fusion (LiDAR, IMU, encoders)
- Designed AI-based perception and path planning pipelines for autonomous navigation
- Simulated the system in Gazebo and validated kinematics, odometry, and trajectory tracking

EDUCATION

B.S. in Mechatronics Engineering — Mansoura University

SKILLS

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, Deep Learning, Object Detection (YOLO), Image Segmentation, Face Recognition, Edge AI, ONNX, TensorRT, Model Deployment & Optimization

Generative AI / LLM Experience: RLHF, AI Training Data Validation, Prompt Refinement & Optimization

Robotics & Control: ROS 2, SLAM, Navigation Stack, Sensor Fusion, Path Planning

Embedded Systems: ESP32, Raspberry Pi, micro-ROS, C++, RTOS, Communication Protocols

Programming & Tools: Python, C++, Linux (Ubuntu), Git, GitHub

Data & Processing: NumPy, Pandas, Image Processing

COURSES & CERTIFICATIONS

- Artificial Intelligence Diploma — Amit Learning
- Mathematics for Machine Learning — DeepLearning.AI
- Machine Learning Specialization — DeepLearning.AI
- Deep Learning Specialization — DeepLearning.AI
- Embedded Systems Diploma — IMT School

LANGUAGES

Arabic (Native) | English (Advanced)